GATE

# Computer Science & Information Technology

## Computer Networks

DPP: 1

## IP address Subnetting, Supernetting

**Q1** An organization is granted the block 150.36.0.0/16. The administrator wants to create 512 subnets.
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What is the subnet mask?
(A) 255.255.255.128/25
(B) 255.255.255.192/26
(C) 255.255.255.224/27
(D) 255.255.255.240/28

**Q2** What could be the network mask if the direct broadcast address of a network is 168.17.07.255?
(A) 255.255.248.0 (B) 255.255.252.0
(C) 255.255.254.0 (D) All the above

**Q3** The subnet mask for a particular network is 255.255.252.0. Which of the following pairs of IP addresses could belong to this network?
(A) 172.57.88.62 and 172.57.87.233
(B) 10.35.24.2 and 10.35.29.4
(C) 191.203.31.87 and 191.234.31.88
(D) 128.8.129.43 and 128.8.131.42

**Q4** In IP4 addressing format, the number of networks allowed under Class-C address is:
(A) $2^{24}$ (B) $2^{21}$
(C) $2^{8}$ (D) $2^{8}$-2

**Q5** Suppose a subnet 'X' has a subnet mask 255.255.192.0 and a system A has IP 157.106.46.234. Which of the following IPs belongs to the same network A?
(A) 157.106.65.03
(B) 157.106.142.77
(C) Both (A) and (B)
(D) None of these

**Q6** A router uses the following routing table:

| Destination | Mask | Interface |
|---|---|---|
| 144.72.0.0 | 255.255.0.0 | Eth0 |
| 144.72.64.0 | 255.255.224.0 | Eth1 |
| 144.72.68.0 | 255.255.255.0 | Eth2 |
| 144.72.68.64 | 255.255.255.224 | Eth3 |

A packet bearing a destination address 144.72.68.117 arrives at the router on which interface will it be forwarded?
(A) Eth0 (B) Eth1
(C) Eth2 (D) Eth3

**Q7** Let computers A and B have IP addresses 72.195.126.113 and 72.195.126.91, respectively, and both use subnet mask 'N'. Then what is the value of 'N' that should not be used out of the following if both belong to the same network?
(A) 255.255.255.0
(B) 255.255.255.128
(C) 255.255.255.192
(D) 255.255.255.224

**Q8** 127.0.127.195 is a:
(A) Limited Broadcast Address
(B) Direct Broadcast Address
(C) Multicast Address
(D) Loopback Address

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**Q9** How many networks of class B are possible

(A) $2^{32}$ (B) $2^{16}$

(C) $2^{14}$ (D) $2^{7}$

**Q10** In which of the following strategies, bits from HID are chosen in an IP address. (HID means Host ID).

(A) Subnetting (B) Supernetting

(C) NAT (D) None of these

**Q11** 255.255.63.0 is the subnet mask for the network. Which of the following pairs of IP addresses could belong to same network?

(A) 176.64.88.62 and 176.64.87.23.42

(B) 11.45.28.12 and 11.45.99.24

(C) 125.18.130.25 and 125.18.194.46

(D) 193.213.31.67 and 193.213.96.89

**Q12** You would like to set up an office LAN connected to the Internet via a dedicated router. There should be up to 511 hosts with IP addresses connected to the LAN. What is the longest possible netmask for the subnet (the LAN)?

(A) 255.255.248.0 (B) 255.255.252.0

(C) 255.255.240.0 (D) 255.255.0.0

**Q13** How many subnets and number of hosts per subnet is possible?

For a class B network, which has a subnet mask of 255.255.248.0

(A) 30, 1024 (B) 30, 2046

(C) 32, 2046 (D) 126, 512

**Q14** A router has the following (CIDR) entries in the routing table.

| Address/mask | Next hop |
|---|---|
| 138.48.56.0/22 | I-0 |
| 138.48.60.0/22 | I-1 |
| 192.150.48.0/23 | R-1 |
| default | R-2 |

Next hop for a packet with IP address 138.48.63.10

(A) I-0 (B) I-1

(C) R-1 (D) R-2

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# Answer Key

| | | | |
|---|---|---|---|
| Q1 | (A) | Q8 | (D) |
| Q2 | (D) | Q9 | (B) |
| Q3 | (B) | Q10 | (A) |
| Q4 | (B) | Q11 | (C) |
| Q5 | (D) | Q12 | (B) |
| Q6 | (C) | Q13 | (B) |
| Q7 | (D) | Q14 | (D) |

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# Hints & Solutions

**Q1 Text Solution:**

From the Question, 16 bits are for the NID part and 16 bits are for the HID part. We need to create 512 subnets. So we require 9 bits to be borrowed from HID, i.e., 8 bits from the 3rd octet and 1 bit from the 4th octet.

Therefore, the total number of l's in the subnet mask will be 16 (because of the network id part) +9 (because of the subnet part) = 25

Hence the subnet mask will be 255.255.255.128/25

**Q2 Text Solution:**

Given IP address 168.17.07.255 is DBA and its class B. In DBA, the HID part is all 1s

Therefore, 168.17.00000111.11111111

NID HID

For Netmask, the NID part is all 1's and the HID part is all O's

Since the number of bits in NID or HID is not given, we can take any length for NID. So last 11 bits can be HID (bcz DBA contains all 1's in HID). Similarly last 10 bits can be HID, or 9 bits or 8 bits. Depending on the number of bits, the Subnet Mask will vary.

The range of Netmask can be 255.255.248.0 to 255.255.255.0

**Q3 Text Solution:**

Take each option and do bitwise And with the given subnet mask. If you get the same network for both IPs given in an option, that is the answer.

**Q4 Text Solution:**

For class C address, 8 bits are reserved for Host Id, and 24 bits are reserved for Network Id. But out of 24 bits, the first 3 bits are fixed as 110; hence, the total number of networks possible

**Q5 Text Solution:**

Take the Given IP and given SM, DO AND WITH both, you will get NID Now, take each option and do bitwise AND with the given subnet mask. If you get the SAME NID for IP'S given in an option, that is the answer.

Here, No IP address belongs to the Subnet X.

**Q6 Text Solution:**

Do Bitwise ANDING with the given IP address and each Mask from the table. Results match with the destination IP which will be the correct interface.

if it matches with more than one destination IP, then takes the highest no. of I's present in the Mask interface.

**Q7 Text Solution:**

Do Bitwise AND with SM (from option) and given IP the correct answer will be which give same NID for both IP

**Q8 Text Solution:**

Loopback Address

**Q9 Text Solution:**

In class B, 16 bits are chosen for network ID and from these 16 bits, two leading bits are fixed, i.e. "10", so 14 bits are remaining to use for Host.

**Q10 Text Solution:**

In subnetting, bits from HID are chosen and used as subnet ID

**Q11 Text Solution:**

To find whether hosts belong to same network or not, we have to find their net id, if net id is same then hosts belong to same network and net id can be find by ANDing subnet mask and IP address.

SM 255.255.63.0 255.255.63.0

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IP 125.18.130.25 125.18.194.46

Anding

.....

NID 125.18.2.0 125.18.2.0

**Q12 Text Solution:**

You need 511+2 (first and last can't be used for hosts) + 1 (for router) IP addresses, hence you need 10 bits at least. The longest netmask is 255.255.252.0. (They asked longest possible mask

**Q13 Text Solution:**

Its class B network therefore first 16 bit is for NID, remaining bits are for SID and HID.

So, No of bits for SID is 5 bits and for HID is 11 bits.

The number of subnets possible is $2^5$ - 2= 30

Number of Host addresses possible is $2^N$ - 2 = 2046 (N=11)

**Q14 Text Solution:**

Mask for I-0-255.255.252.0, 1-1-255.255.252.0, R-1-255.255.254.0 Given IP = 138.48.52.2,

Do logical AND with Mask, result NID will not be matched to any of the hope, Goes to Default.

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